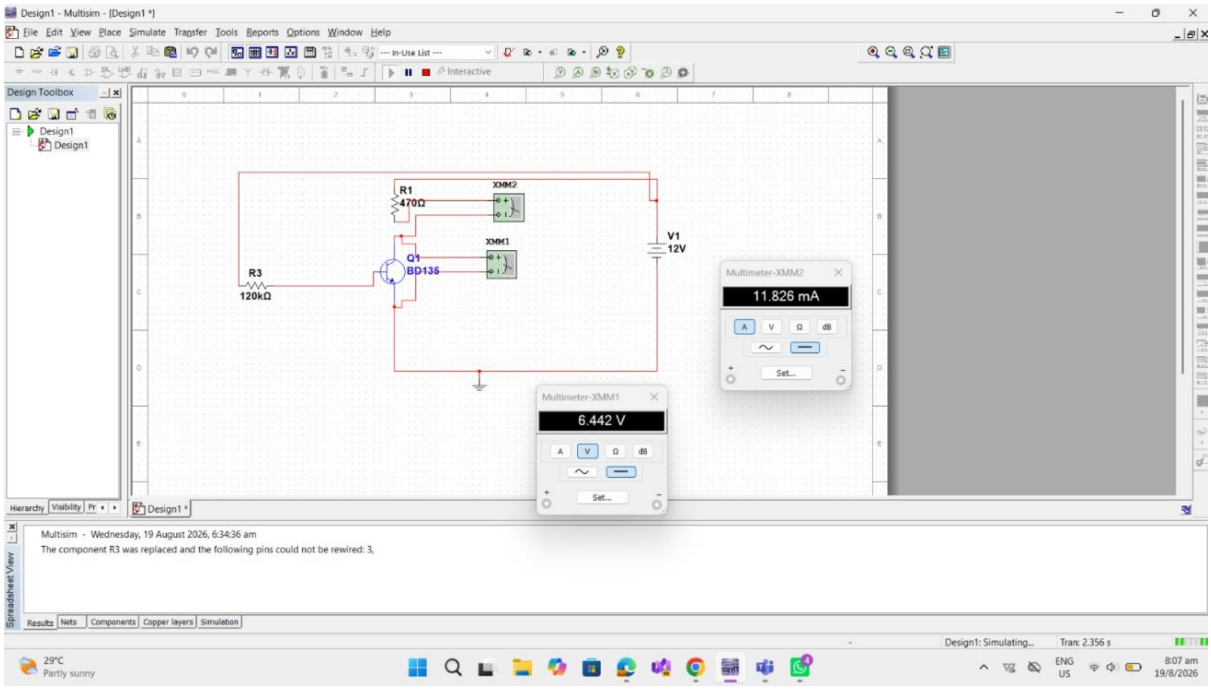
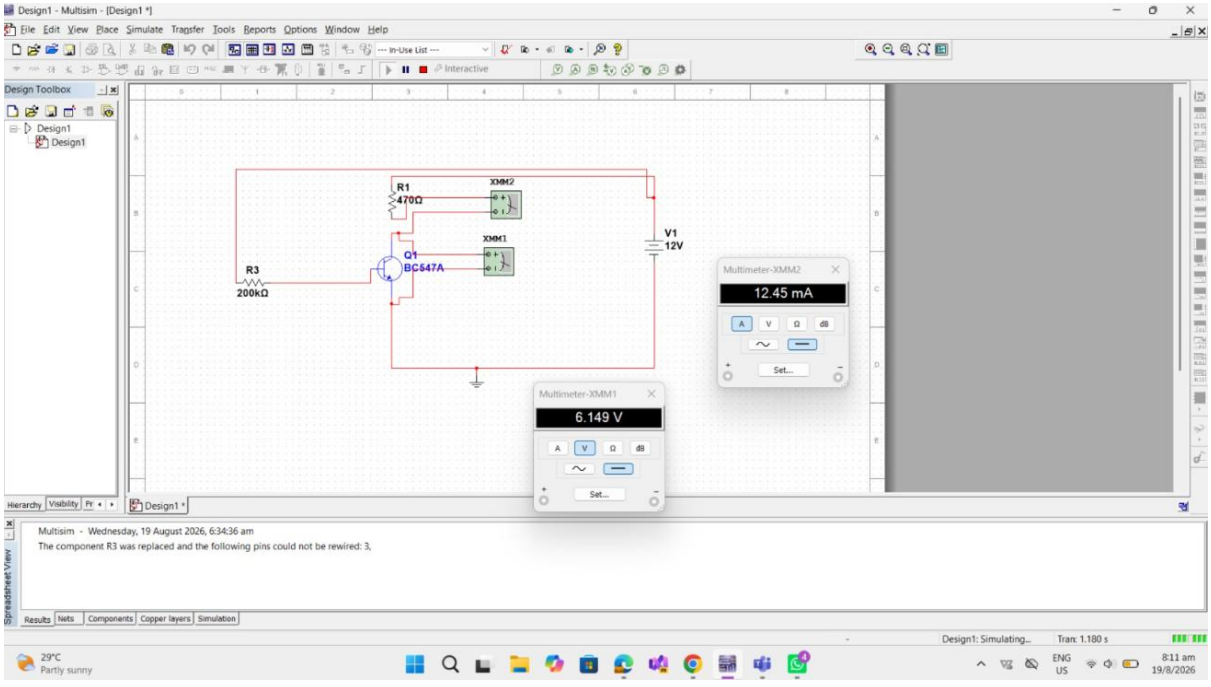
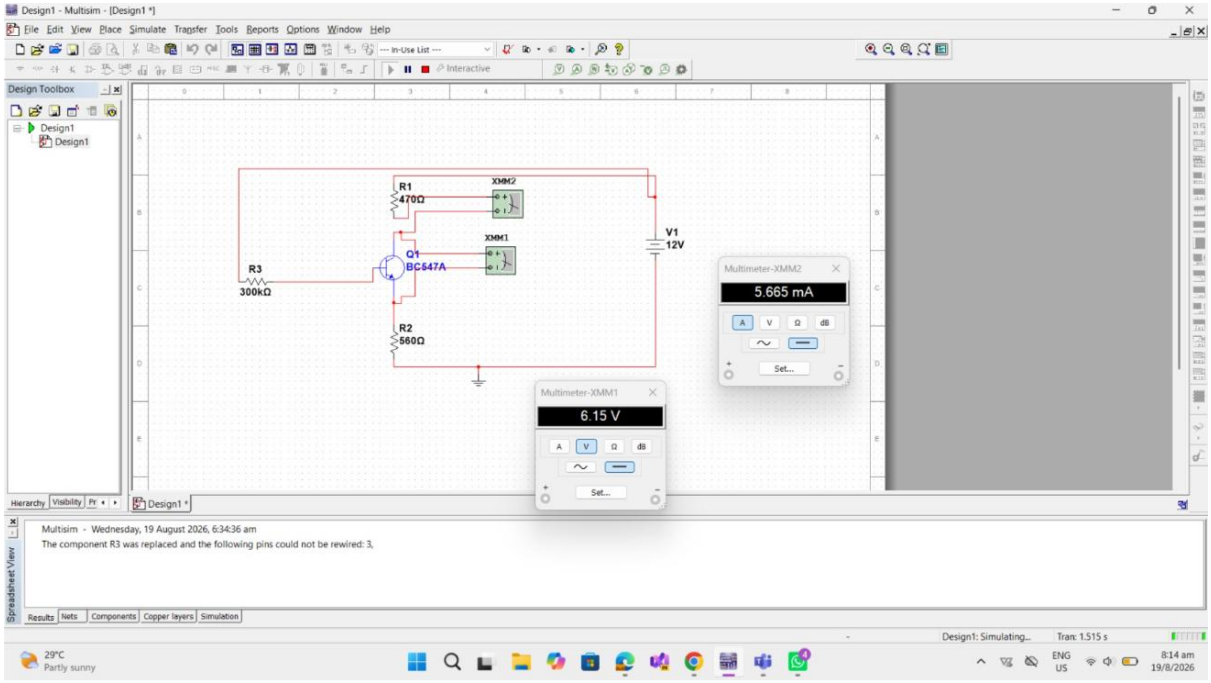
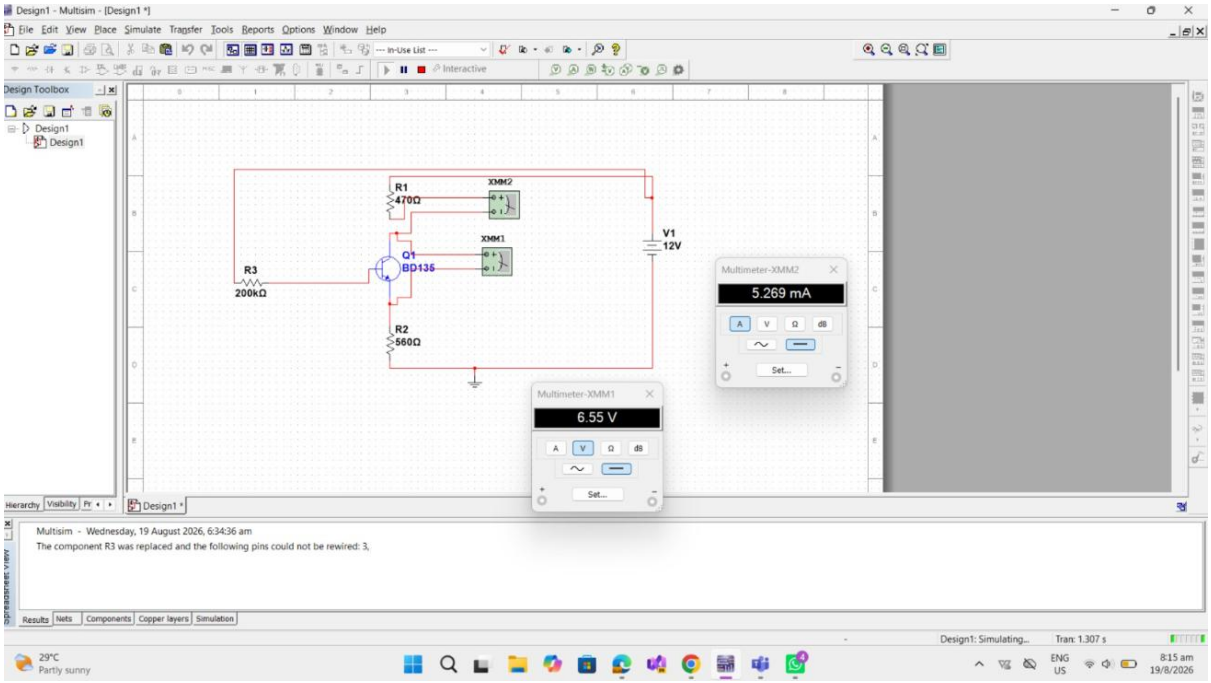
Fixed Bias


Design1 - Multisim - [Design1 *]
File Edit View Place Simulate Transfer Tools Reports Options Window Help
Design Toolbox
Design1
Design1
R1
470Ω
XMM2
XMM1
Q1
BD135
R3
120kΩ
V1
12V
Multimeter-XMM2
11.826 mA
Multimeter-XMM1
6.442 V
Set...
Design1 *
Multisim - Wednesday, 19 August 2026, 6:34:36 am
The component R3 was replaced and the following pins could not be rewired: 3,
Results Nets Components Copper layers Simulation
Design1: Simulating...
Tran: 2.356 s
29°C
Partly sunny
ENG
US
8:07 am
19/8/2026

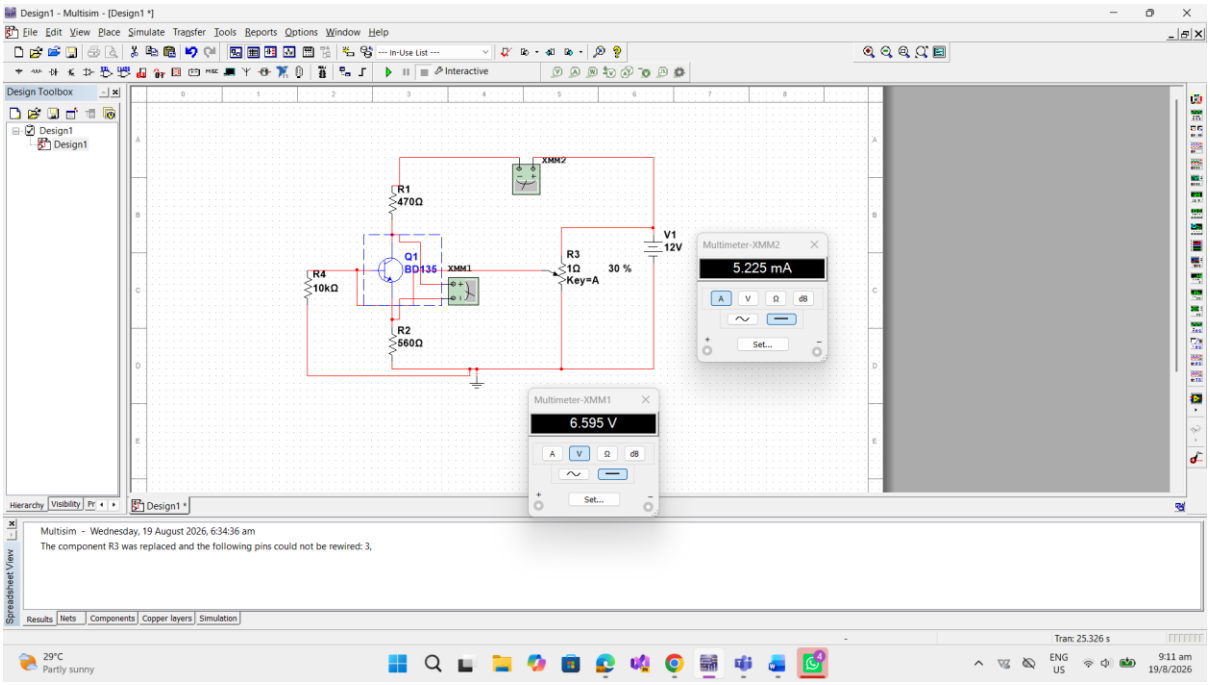

Design1 - Multisim - [Design1 *]
File Edit View Place Simulate Transfer Tools Reports Options Window Help
Design Toolbox
Design1
Design1
R1
470Ω
XMM2
XMM1
Q1
BC547A
R3
200kΩ
V1
12V
Multimeter-XMM2
12.45 mA
Multimeter-XMM1
6.149 V
Set...
Design1 *
Multisim - Wednesday, 19 August 2026, 6:34:36 am
The component R3 was replaced and the following pins could not be rewired: 3,
Results Nets Components Copper layers Simulation
Design1: Simulating...
Tran: 1.180 s
29°C
Partly sunny
ENG
US
8:11 am
19/8/2026

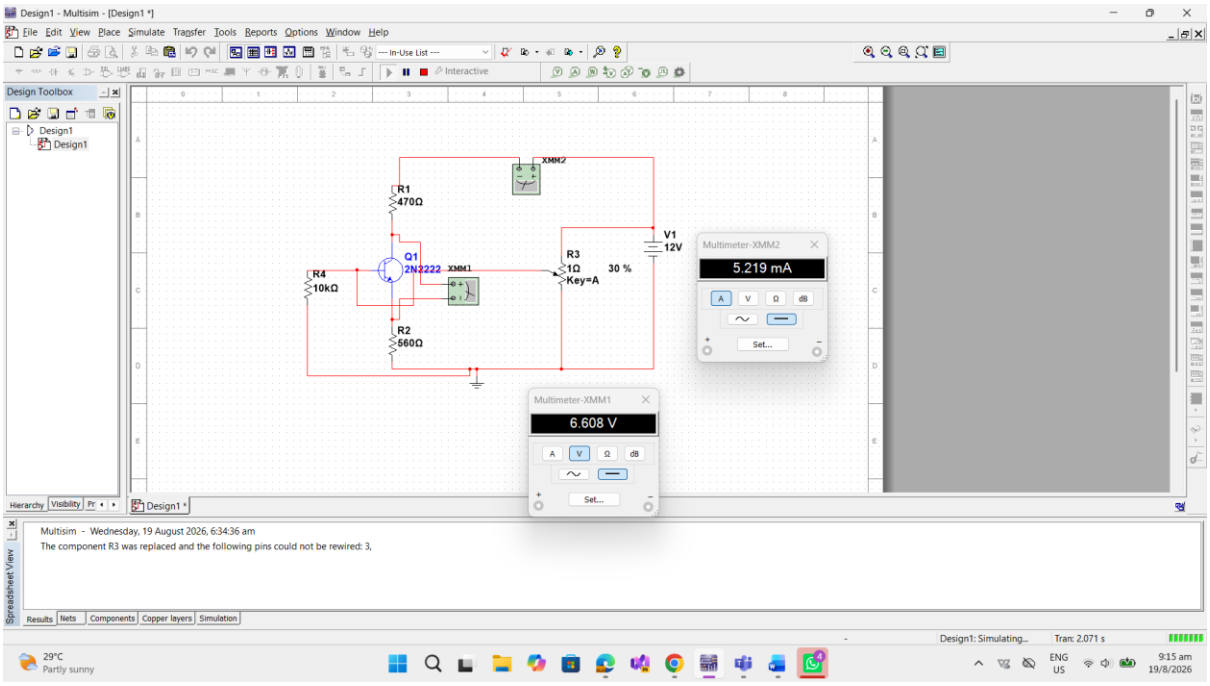
Self-Bias


Design1 - Multisim - [Design1 *]
File Edit View Place Simulate Transfer Tools Reports Options Window Help
Design Toolbox
Design1
Design1
R1
470Ω
XMM2
XMM1
Q1
BC547A
R3
300kΩ
R2
560Ω
V1
12V
Multimeter-XMM2
5.665 mA
Multimeter-XMM1
6.15 V
Set...
Design1 *
Multisim - Wednesday, 19 August 2026, 6:34:36 am
The component R3 was replaced and the following pins could not be rewired: 3,
Results Nets Components Copper layers Simulation
Design1: Simulating...
Tran: 1.515 s
29°C
Partly sunny
ENG
US
8:14 am
19/8/2026

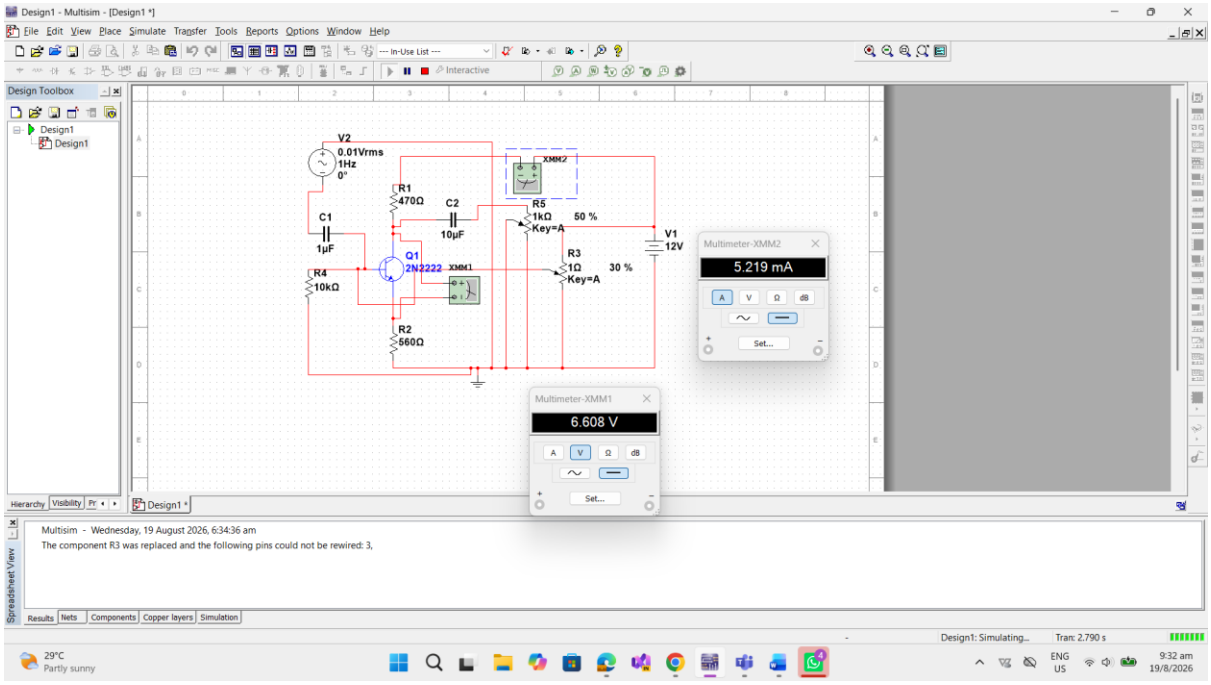

Design1 - Multisim - [Design1 *]
File Edit View Place Simulate Transfer Tools Reports Options Window Help
Design Toolbox
Design1
Design1
R1
470Ω
XMM2
XMM1
Q1
BD135
R3
200kΩ
R2
560Ω
V1
12V
Multimeter-XMM2
5.269 mA
Multimeter-XMM1
6.55 V
Set...
Design1 *
Multisim - Wednesday, 19 August 2026, 6:34:36 am
The component R3 was replaced and the following pins could not be rewired: 3,
Results Nets Components Copper layers Simulation
Design1: Simulating...
Tran: 1.307 s
29°C
Partly sunny
ENG
US
8:15 am
19/8/2026

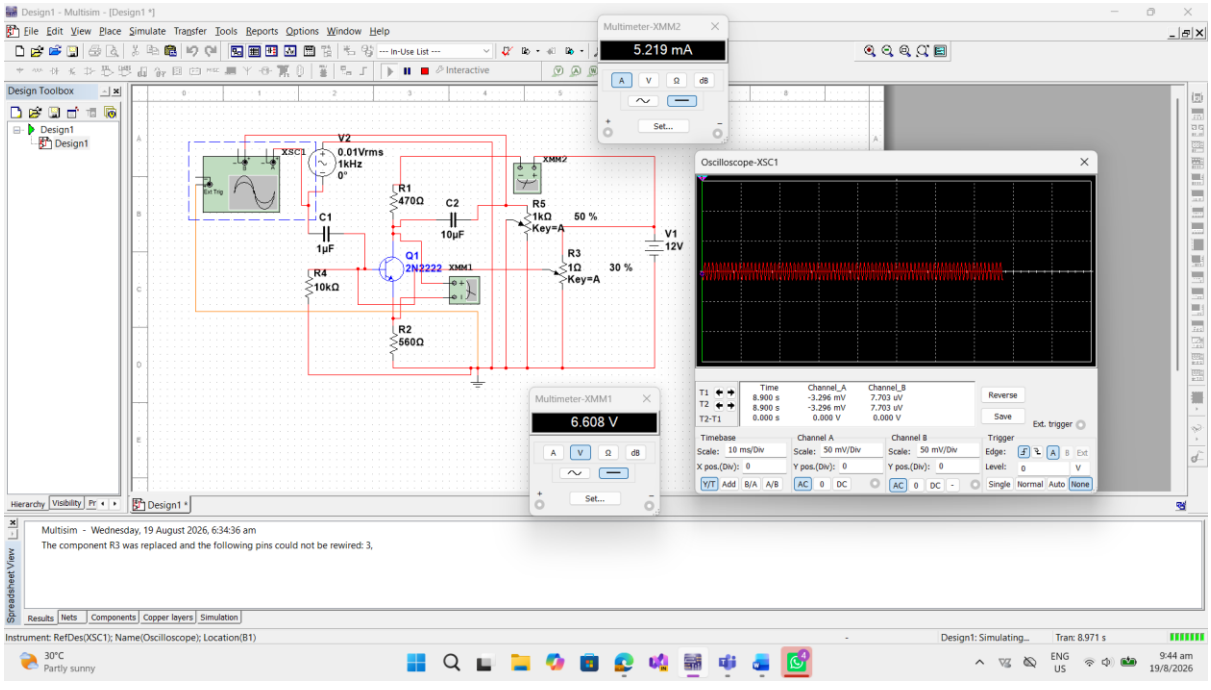
Voltage Divider





Epx 09


Design1 - Multisim - [Design1 *]
File Edit View Place Simulate Transfer Tools Reports Options Window Help
Design Toolbox
V2
0.01Vrms
1Hz
0°
R1
470Ω
C2
10µF
C1
1µF
R5
1kΩ
50 %
Key=A
V1
12V
Q1
2N2222
XMM1
R3
1Ω
30 %
Key=A
R4
10kΩ
R2
560Ω
XMM2
Multimeter-XMM2
5.219 mA
Multimeter-XMM1
6.608 V
Set...
Design1 *
Multisim - Wednesday, 19 August 2026, 6:34:36 am
The component R3 was replaced and the following pins could not be rewired: 3,
Results Nets Components Copper layers Simulation
Design1: Simulating...
Tran: 2.790 s
29°C
Partly sunny
ENG
US
9:32 am
19/8/2026


Design1 - Multisim - [Design1 *]
File Edit View Place Simulate Transfer Tools Reports Options Window Help
Design Toolbox
XSC1
V2
0.01Vrms
1kHz
0°
R1
470Ω
C2
10µF
C1
1µF
R5
1kΩ
50 %
Key=A
V1
12V
Q1
2N2222
XMM1
R3
1Ω
30 %
Key=A
R4
10kΩ
R2
560Ω
XMM2
Multimeter-XMM2
5.219 mA
Multimeter-XMM1
6.608 V
Oscilloscope-XSC1
Set...
Design1 *
Multisim - Wednesday, 19 August 2026, 6:34:36 am
The component R3 was replaced and the following pins could not be rewired: 3,
Results Nets Components Copper layers Simulation
Instrument: RefDes(XSC1); Name(Oscilloscope); Location(B1)
Design1: Simulating...
Tran: 8.971 s
30°C
Partly sunny
ENG
US
9:44 am
19/8/2026